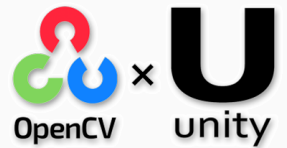


CV VTuber Example 1.1.4

CV VTuber Example



About CV VTuber Example

CVVTuberExample (Computer **V**ision **V**irtual You**T**uber Example) is an **example project** of controlling **3D humanoid models** using **WebCamTexture** or **video files**. You can control the **head orientation** and **facial expressions** of 3D humanoid models using a camera or video frame.

The execution of this asset is required [OpenCV for Unity](#) and [Dlib FaceLandmark Detector](#).



Our Asset Features



Cross Platform Support

- Mobile: iOS & Android

- **Desktop: Windows & macOS & Linux** Standalone support
- **Web: WebGL** support
- **ChromeOS: ChromeOS** support
- **XR: visionOS** support (beta) & **Windows10 UWP** support
- **Editor:** Support for preview in the **Editor**



Supported 3D Models

- **Mecanim Humanoid** - General Unity humanoid models
- **Unity-chan! Model** - Popular Unity mascot character
- **VRM Model** - VRM format models (1.x)
- **Live2DCubism5 Model** - Live2D Cubism models

When creating and publishing an application using this asset, please check the licenses and terms of use of SDK or 3D model.



Core Technology

The head orientation and face expression are controlled by the following procedure:

1. **WebCamTextureMatSourceGetter** - Convert WebCamTexture to OpenCV's Mat class
2. **VideoCaptureMatSourceGetter** - Convert video frames loaded with VideoCapture to OpenCV's Mat class
3. **MultiSourceMatSourceGetter** - Converts frames from various sources to OpenCV's Mat class using MultiSource2MatHelper
4. **DlibFaceLandmarkGetter** - Detect face landmark points from OpenCV's Mat class
5. **DlibHeadRotationGetter** - Estimate head orientation from face landmark points
6. **HeadRotationController** - Control the head orientation of the 3D model using the estimated head orientation
7. **HeadLookAtIKController** - Set Animator.SetLookAtPosition() method using the estimated head orientation
8. **DlibFaceBlendShapeController** - Control the face BlendShape of the 3D model using the face landmark point



Technical Implementation

- **OpenCV Integration:** Built on **OpenCV for Unity** for computer vision capabilities
- **Dlib FaceLandmark Detector:** Uses **Dlib FaceLandmark Detector** for precise facial landmark detection
- **Real-time Processing:** Optimized for real-time face tracking and animation
- **Easy Integration:** Designed to work seamlessly with Unity's development environment

Example Scenes

Basic Examples

- **MultiSource CV VTuber Example** - Control 3D models using video files and webcam feed

Advanced Examples (require add-ons setup)

- **UnityChan CV VTuber Example** - Unity-chan! model integration
 - **VRM10 CV VTuber Example** - VRM 1.0 format support
 - **Live2DCubism5 CV VTuber Example** - Live2D Cubism 5 integration
-

Quick Links

- **Official Site** - Main product page
 - **Example Code** - GitHub repository
 - **Android Demo** - Google Play Store demo
 - **WebGL Demo** - Online WebGL demo
-

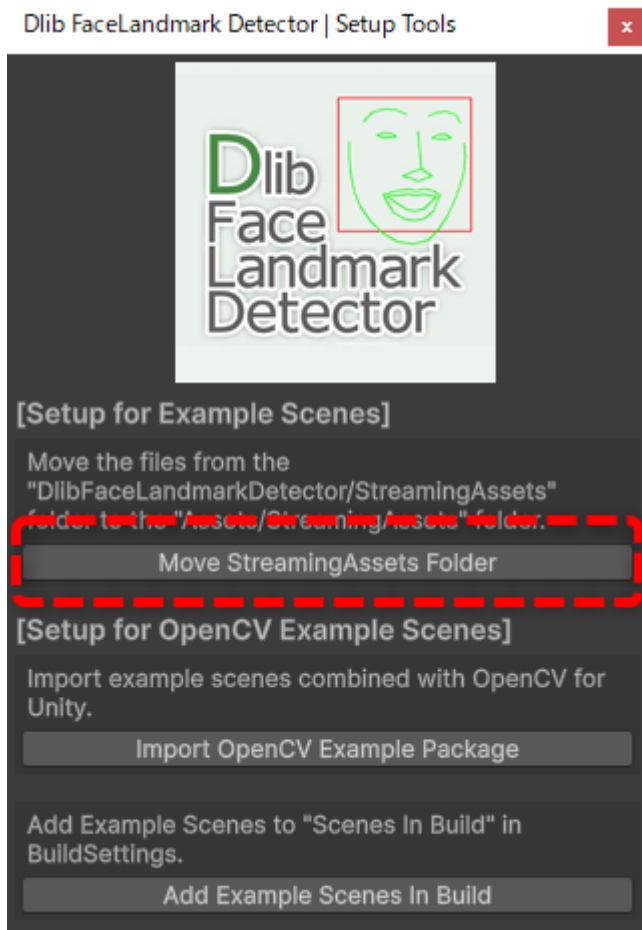
System Requirements

Platform	Requirement
Windows Standalone & Editor	Windows 8 or later
Mac Standalone & Editor	macOS 10.13 or later
Linux Standalone & Editor	Ubuntu 18.04 or later
Android	API level 24 or later
iOS	iOS Version 12.0 or later
visionOS (beta)	visionOS 1 or later

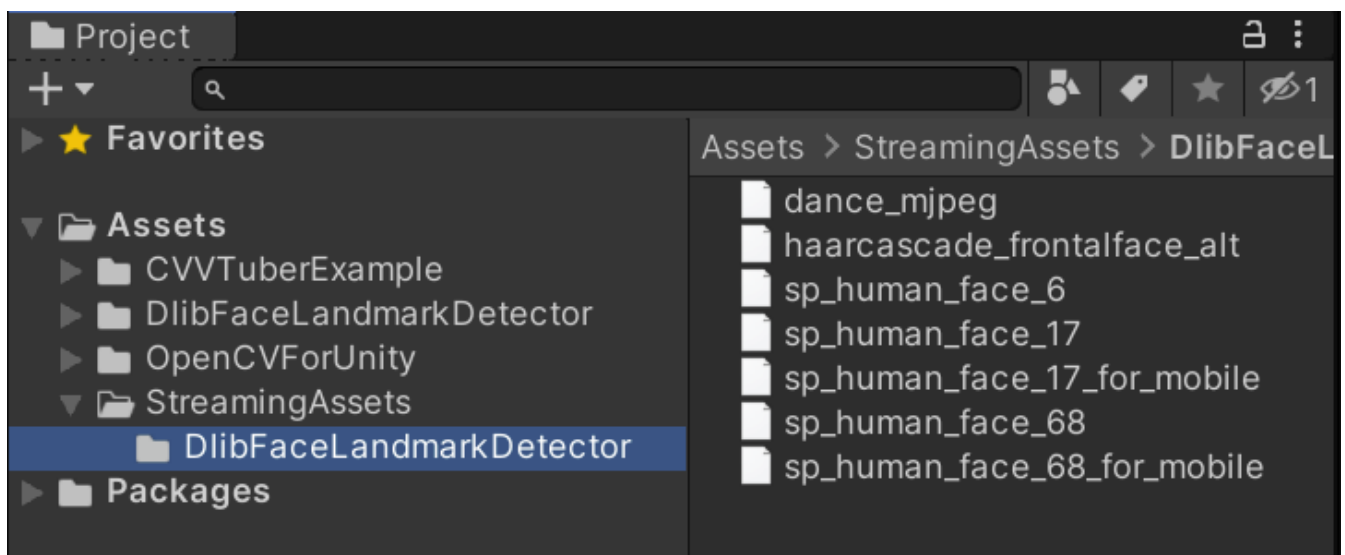
Setup Guide for Getting Started

Quick Setup Procedure for Basic Examples

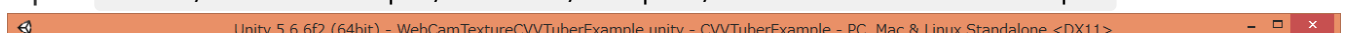
1. Import **OpenCV for Unity** from the Package Manager .
2. Import **Dlib FaceLandmark Detector** from the Package Manager .
3. Import **CVVTuberExample** from the Package Manager .
4. Select the menu item **Tools > Dlib FaceLandmark Detector > Open Setup Tools** to open the Setup Tools window. Click the **Move StreamingAssets Folder** button.

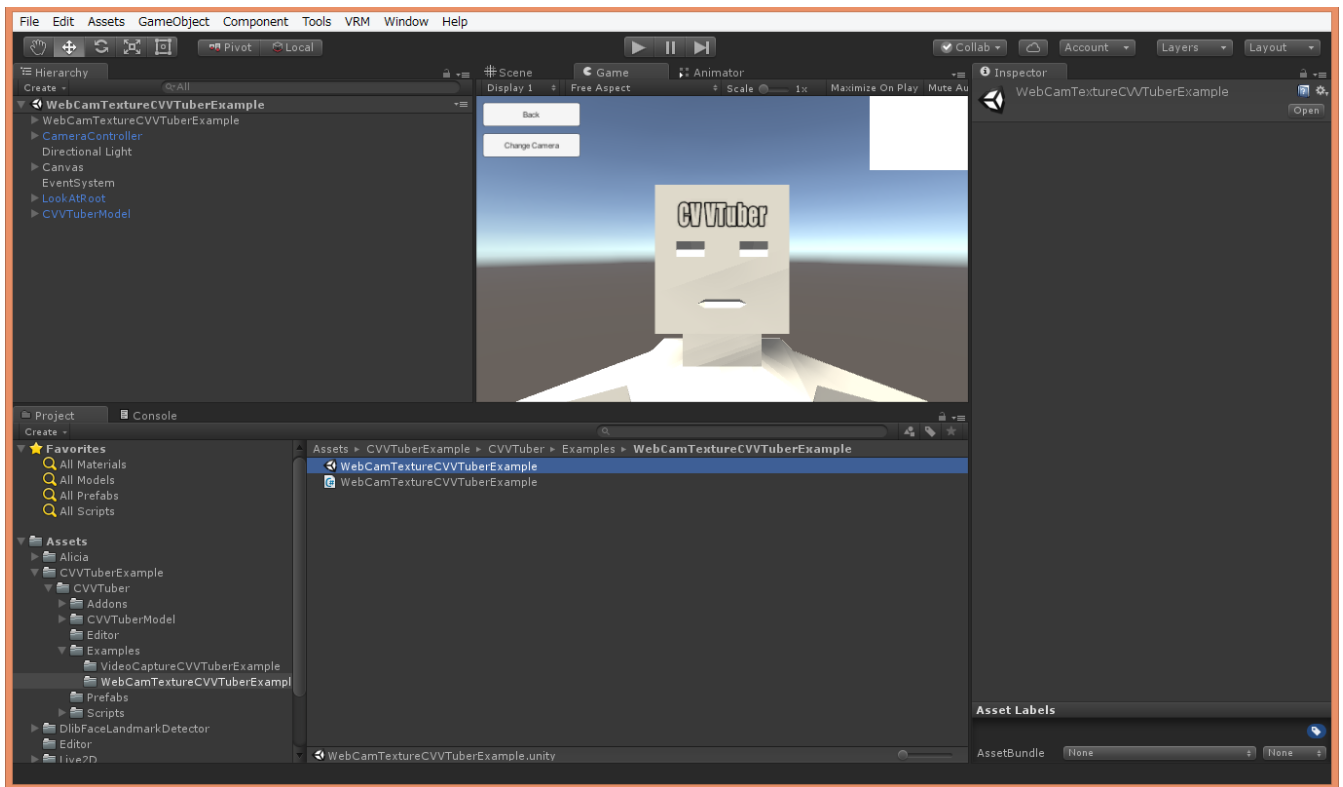


The following files are the only files required for this Example, so other files may be deleted.

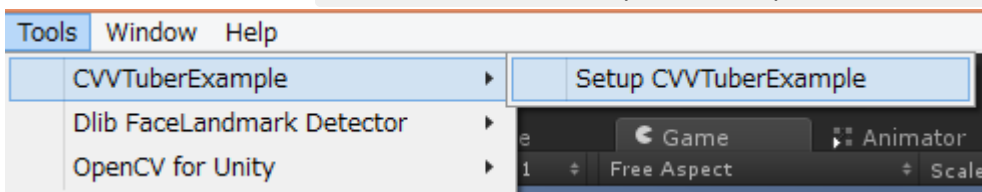


5. Open **Assets/CVVTuberExample/CVVTuber/Examples/MultiSourceCVVTuberExample** scene.

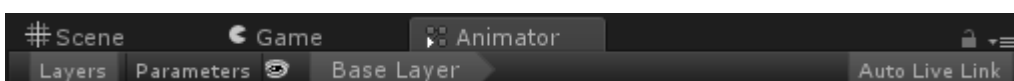
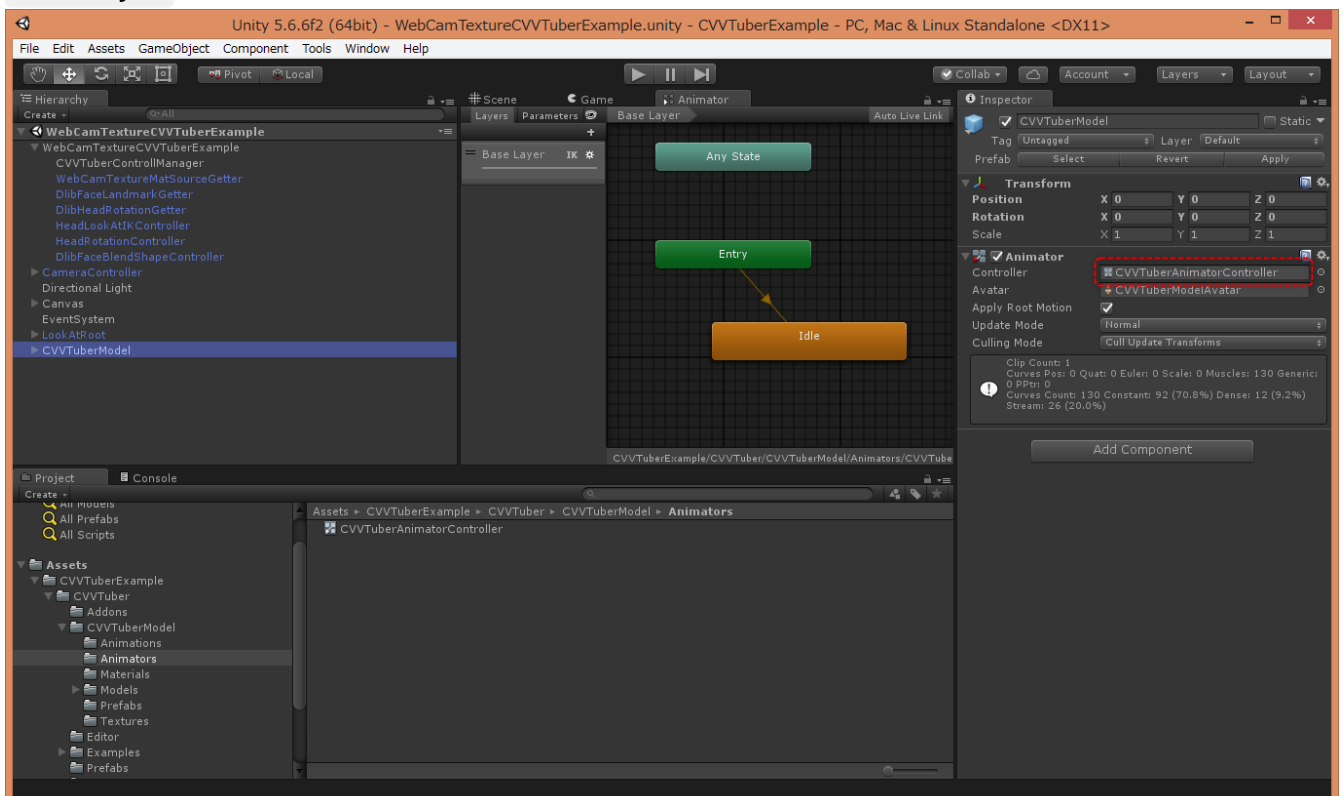


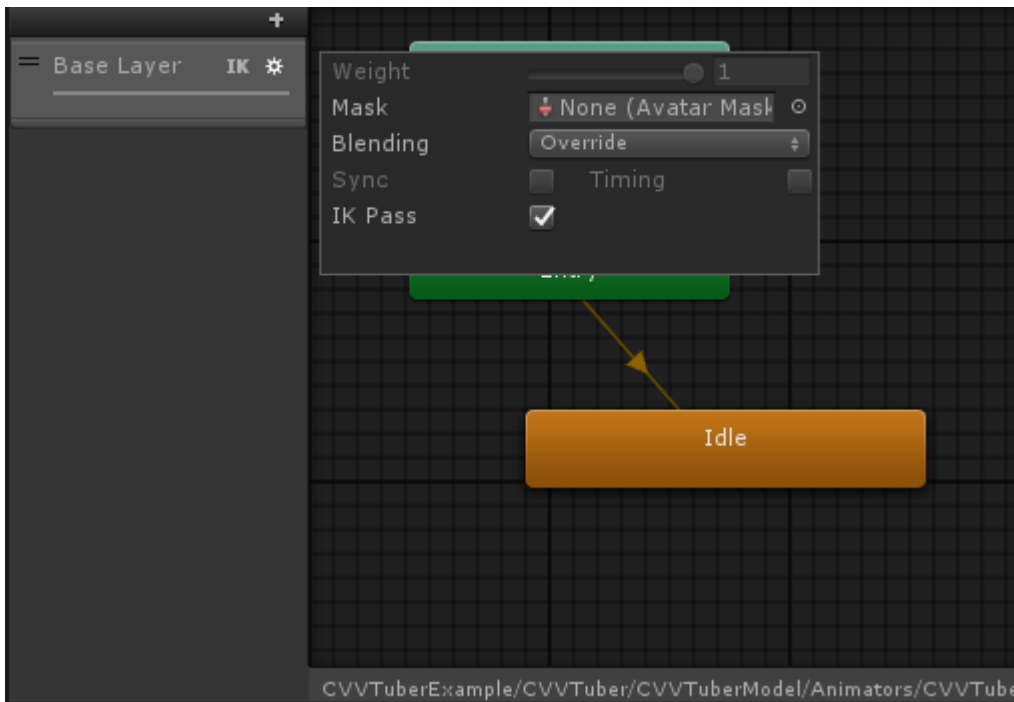


6. Select the menu item **Tools > CVVTuberExample > Setup CVVTuberExample**.

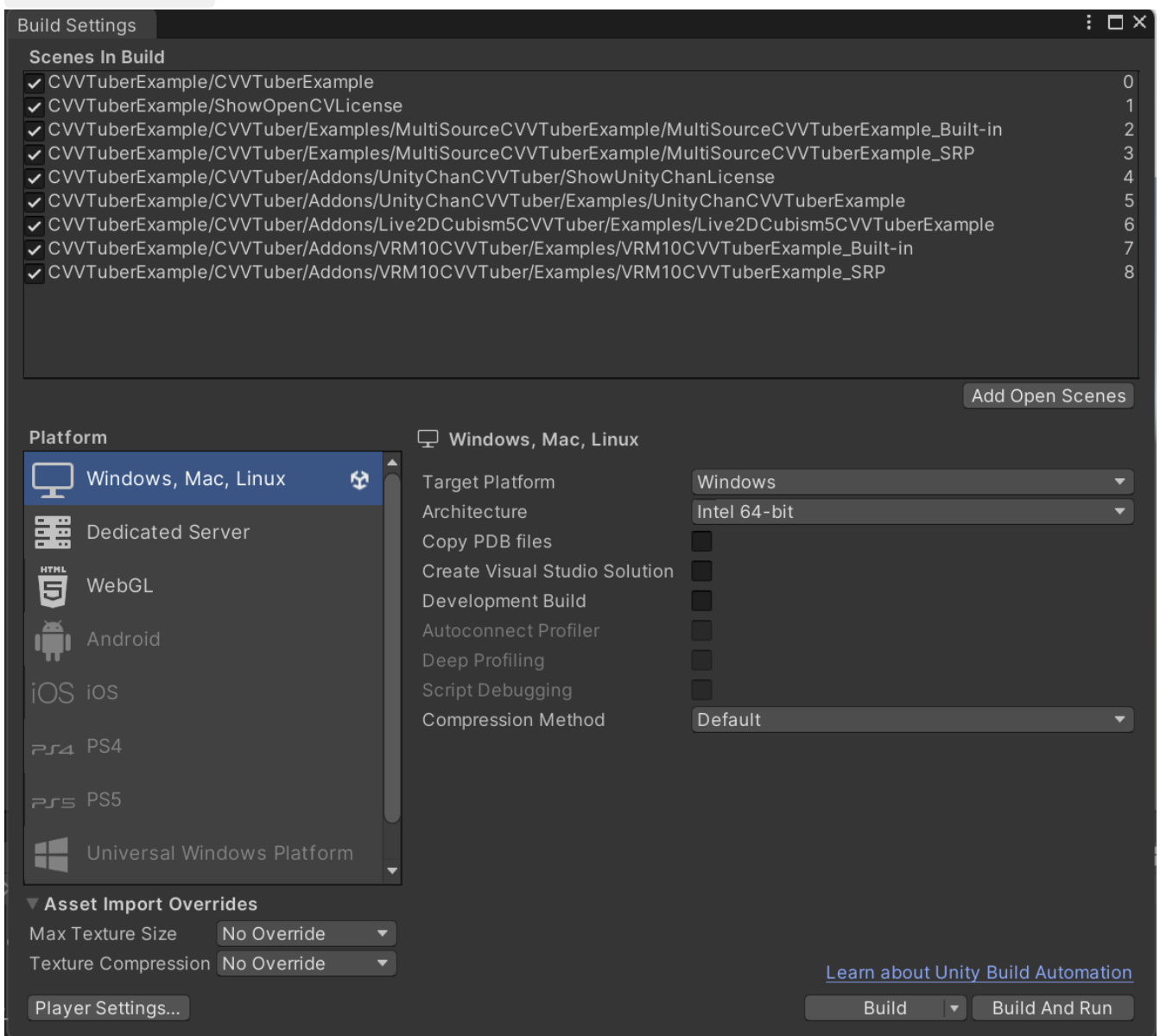


7. Click on **CVVTuberAnimatorController** to open the **Animator** window. Enable **IK Pass** flag of **Base Layer**.

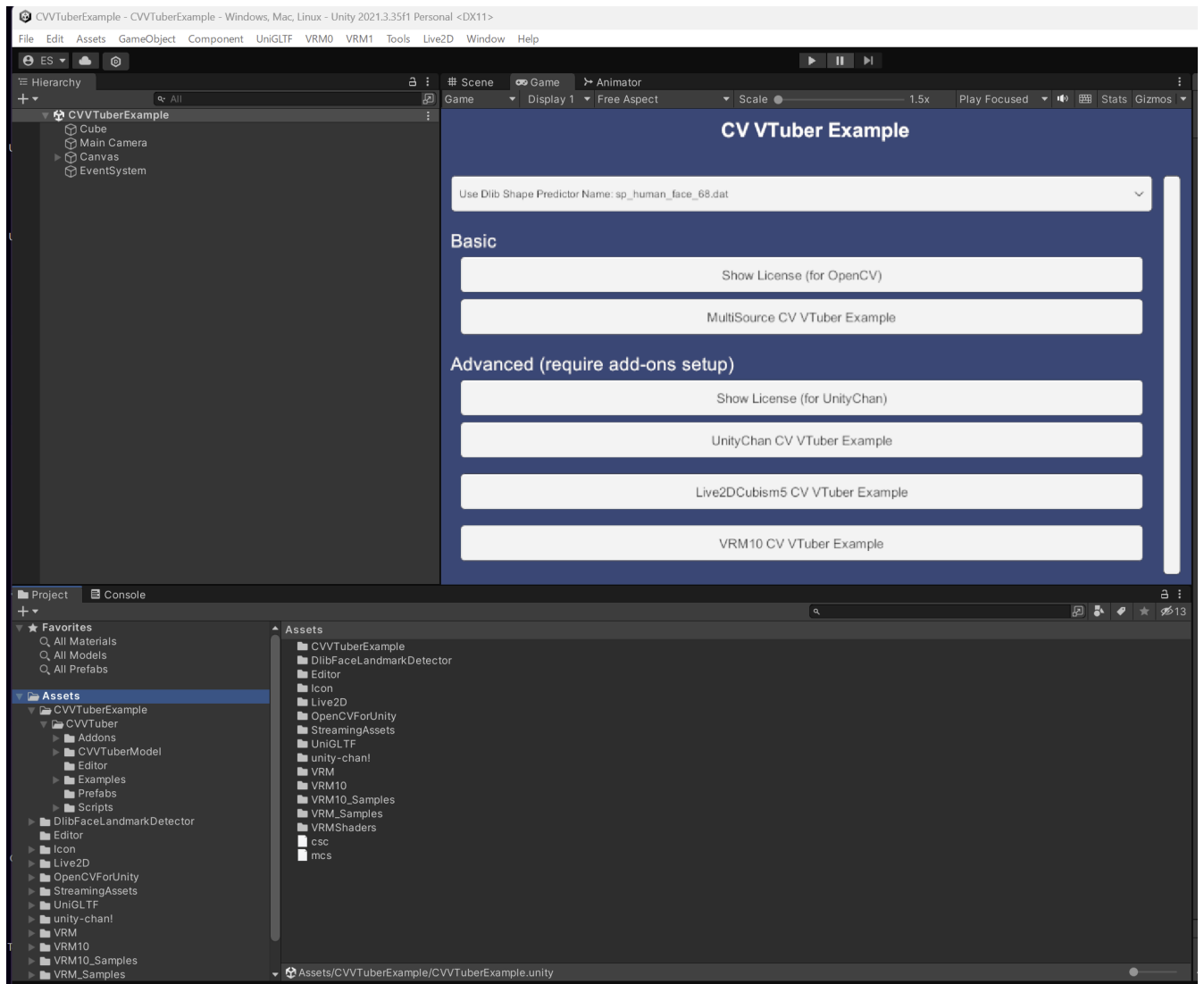




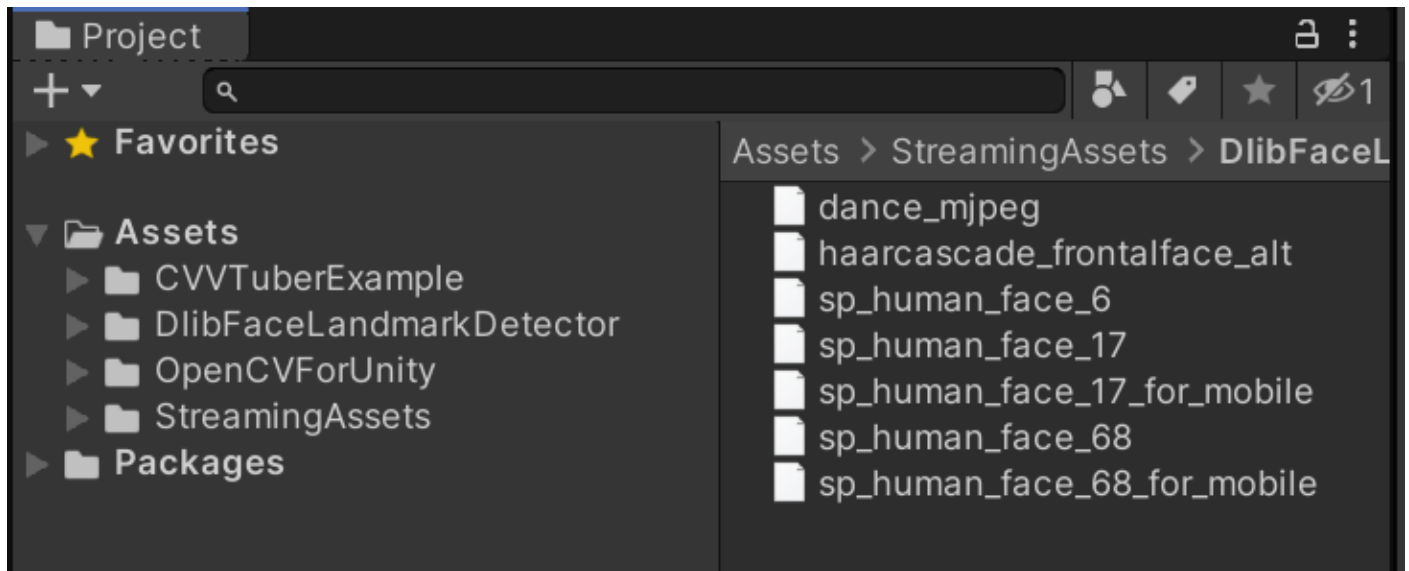
8. Add all of the `***.unity` files in the `CVVTuberExample` folder to `Build Settings` → `Scenes In Build`.



9. Run the `CVVTuberExample` scene.



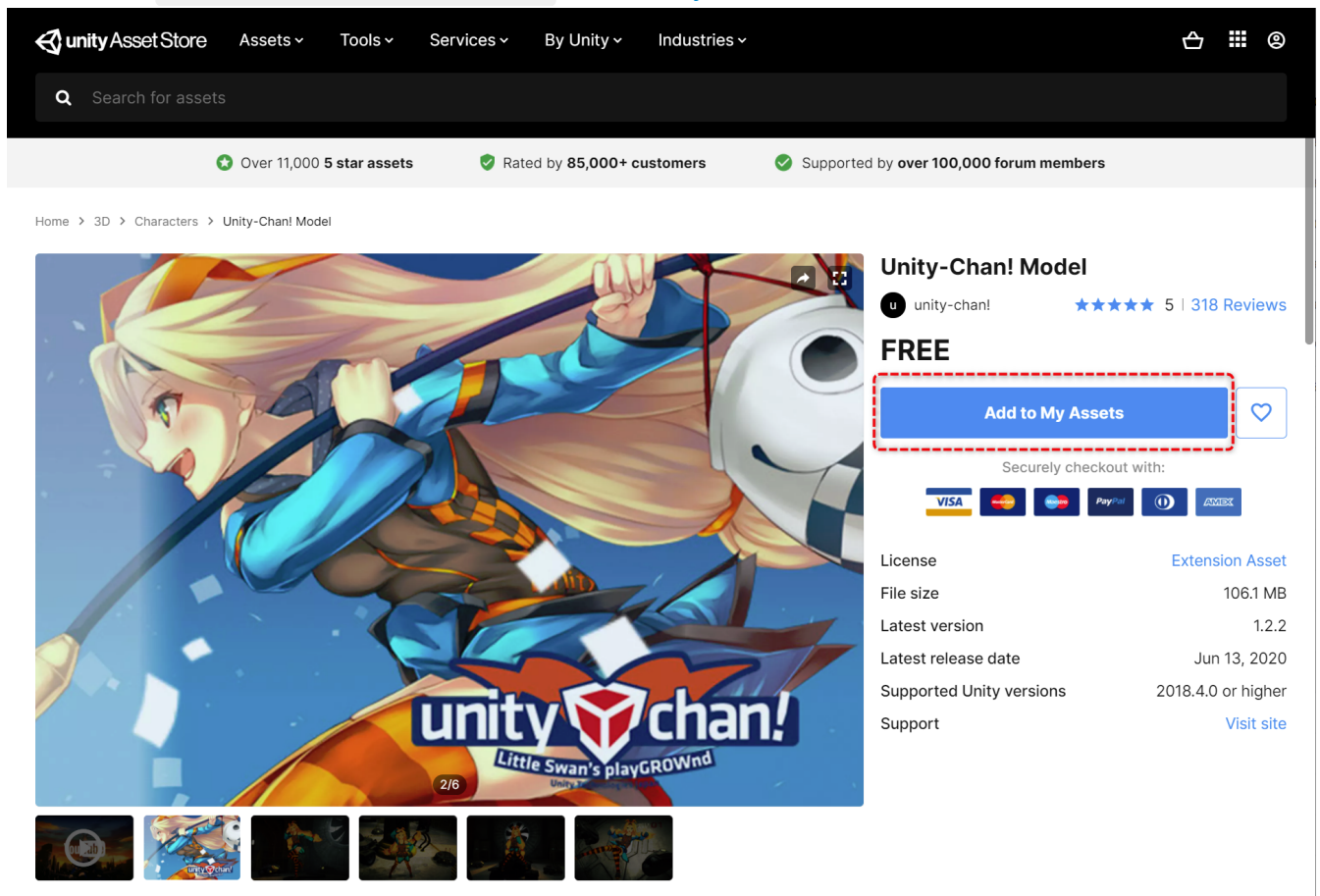
Screenshot after the Setup



In this example, a model set up with a general `Mecanim Humanoid` is also available.

Quick Setup Procedure for UnityChan CV VTuber Example

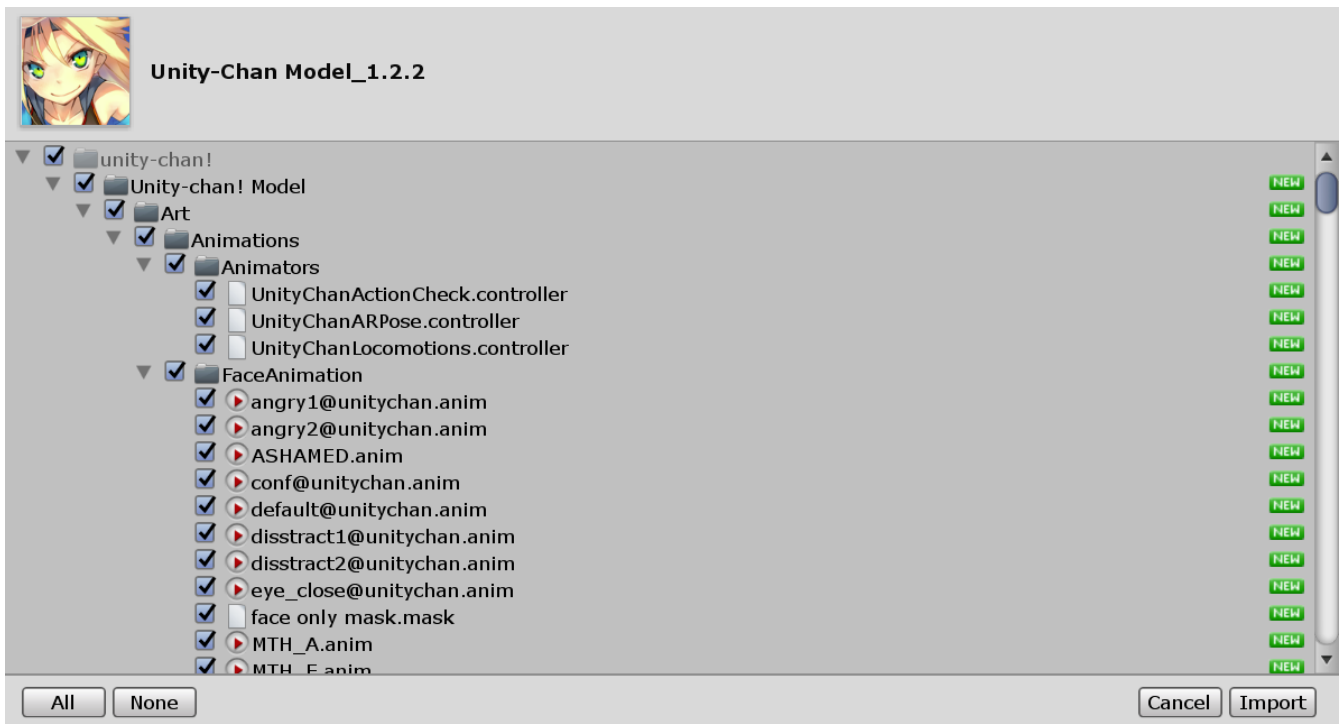
1. Download `Unity-Chan! Model ver1.2.2` from [Unity's AssetStore](#).



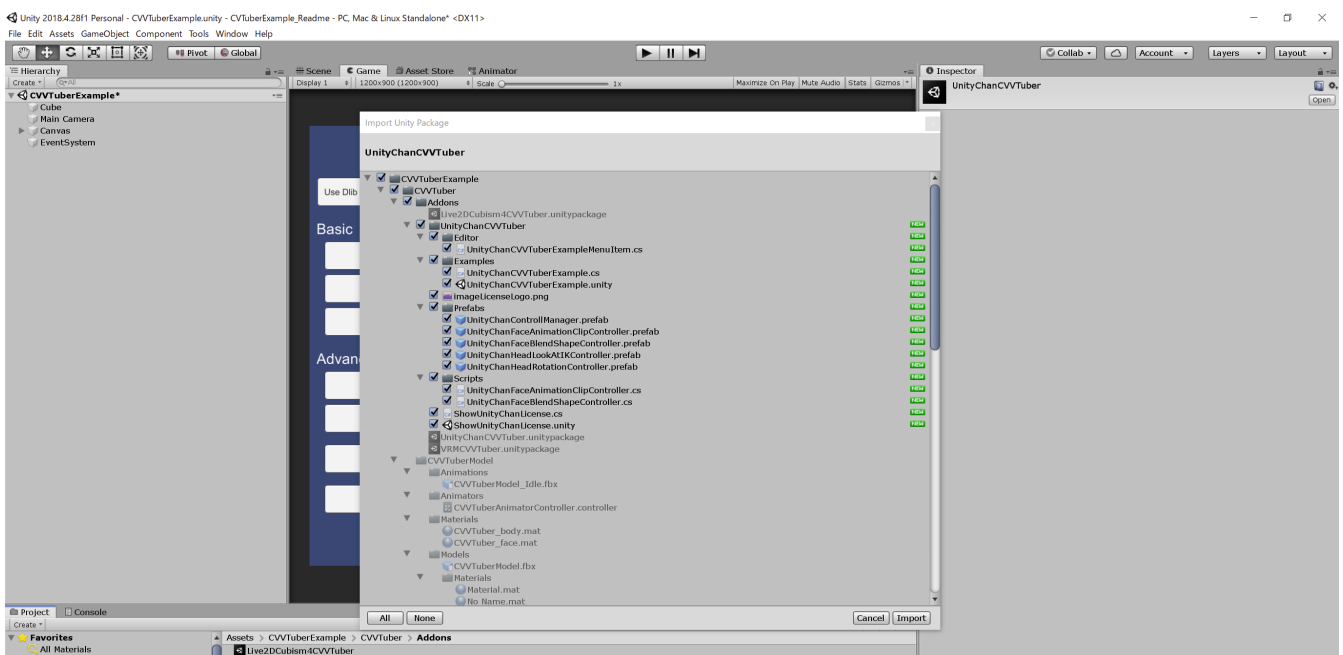
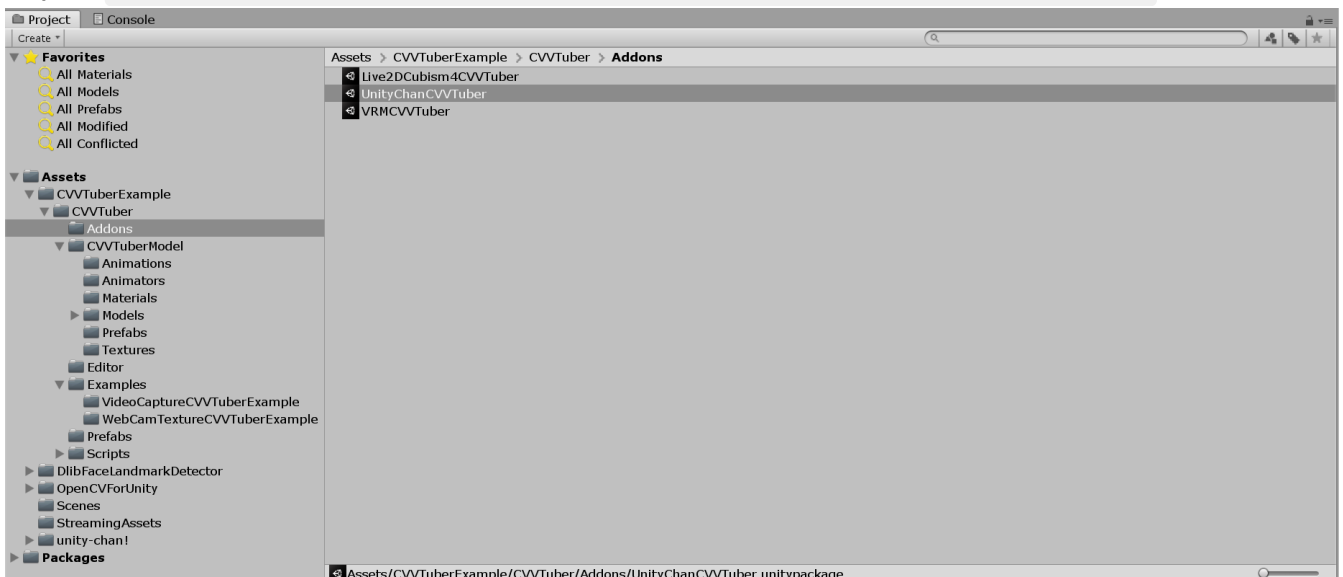
2. Import `Unity-Chan! Model`.

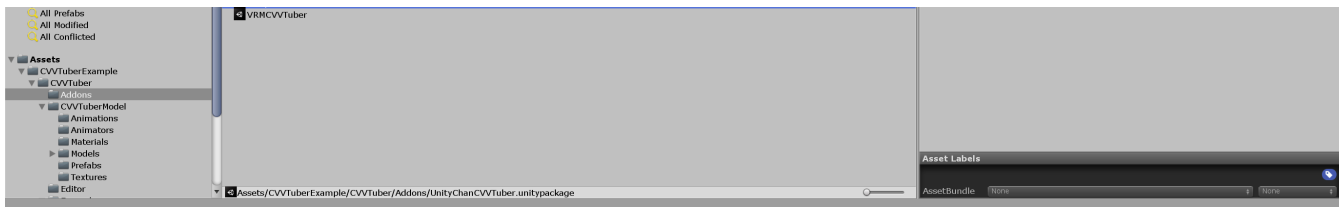
Import Unity Package





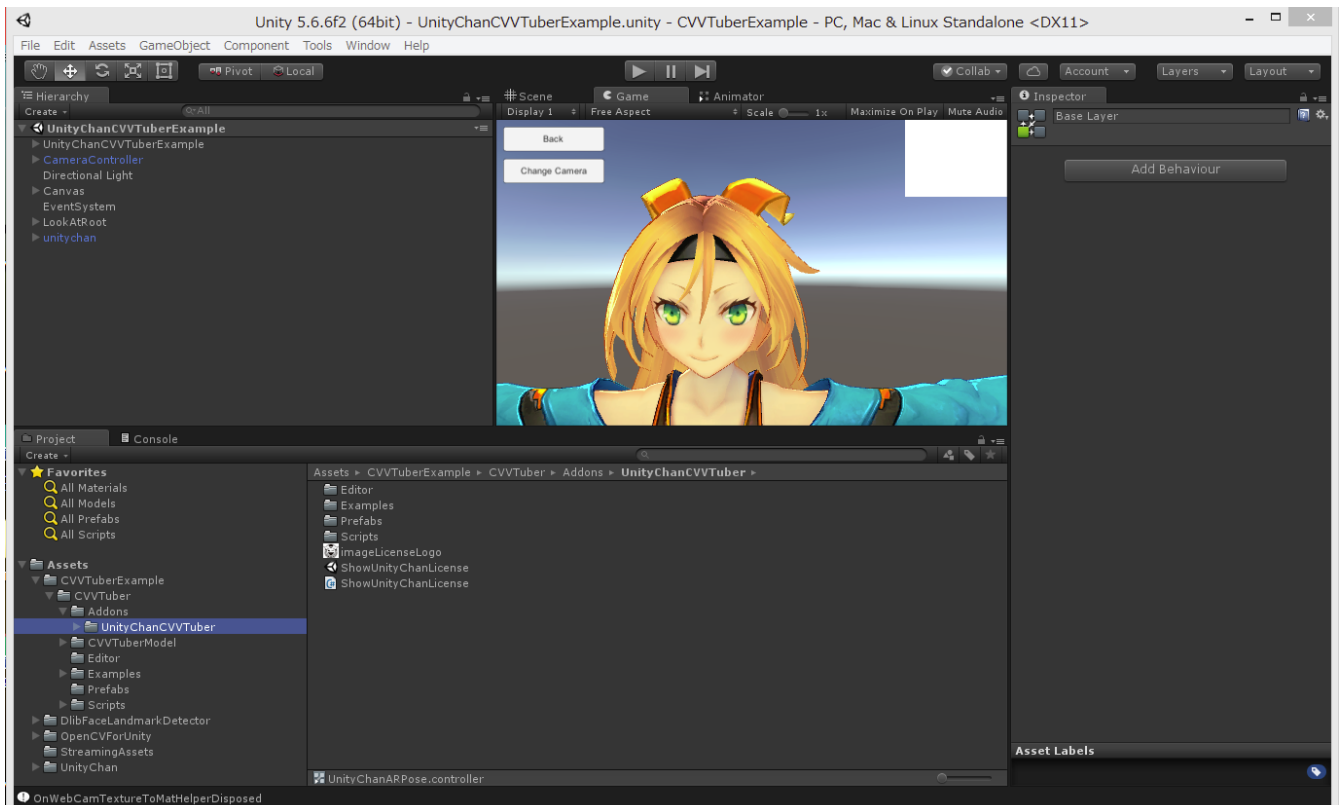
3. Import Assets/CVVTuberExample/CVVTuber/Addons/UnityChanCVVTuber.unitypackage .



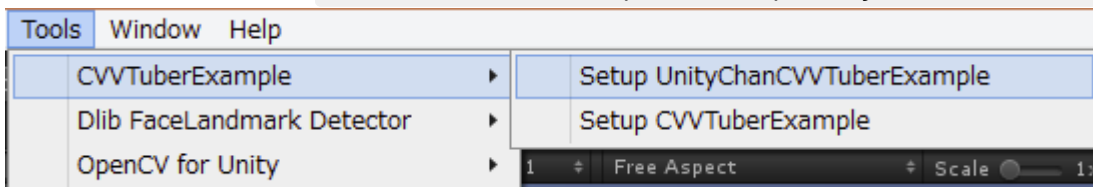


4. Open

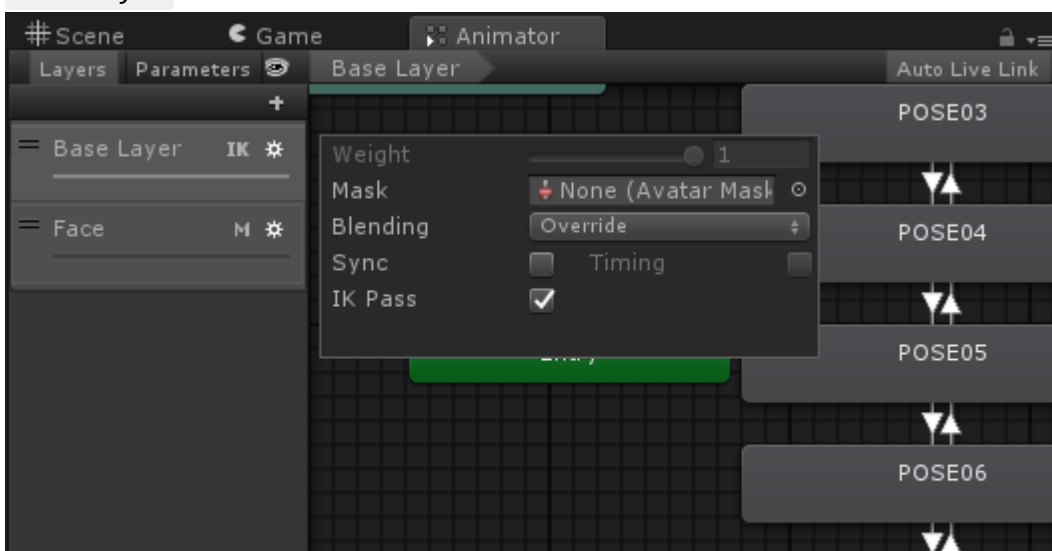
Assets/UnityChanCVVTuber/Examples/UnityChanCVVTuberExample scene.

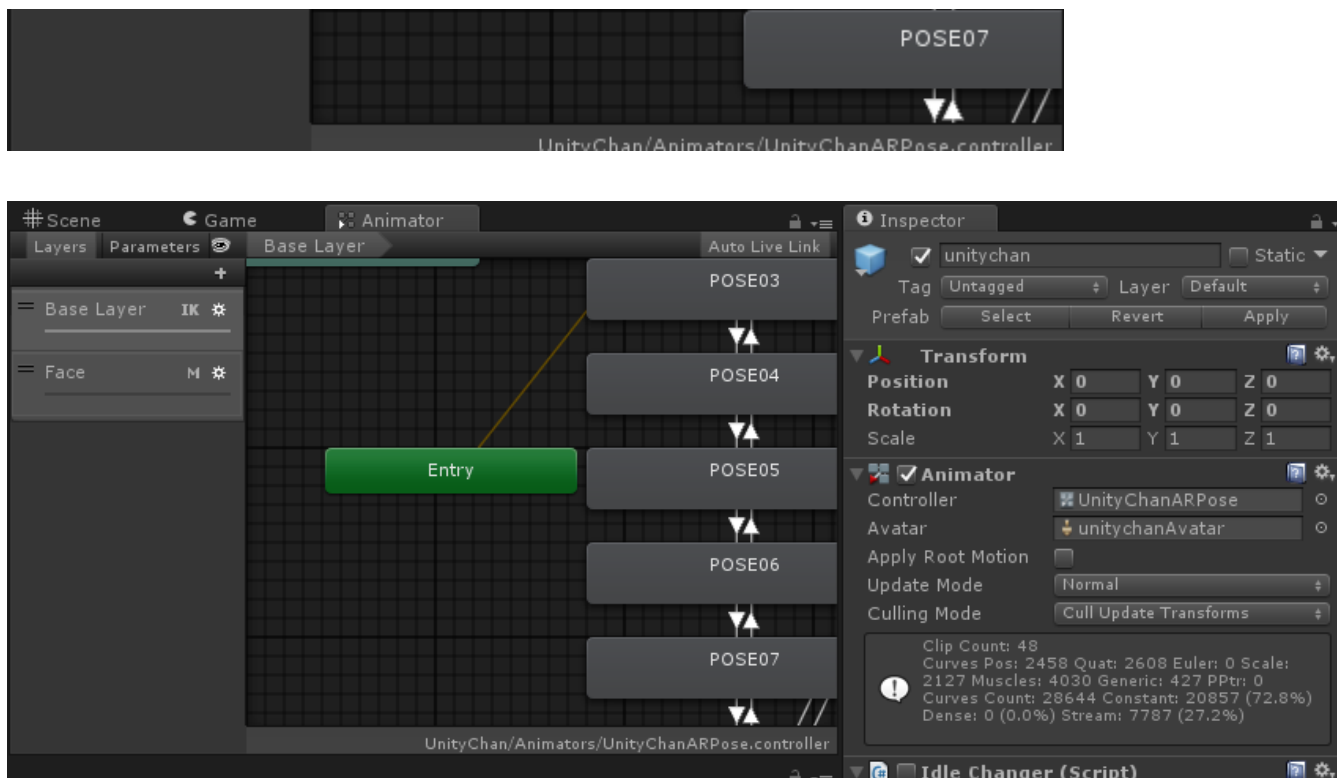


5. Select the menu item Tools > CVVTuberExample > Setup UnityChanCVVTuberExample .



6. Click on CVVTuberAnimatorController to open the Animator window. Enable IK Pass flag of Base Layer .

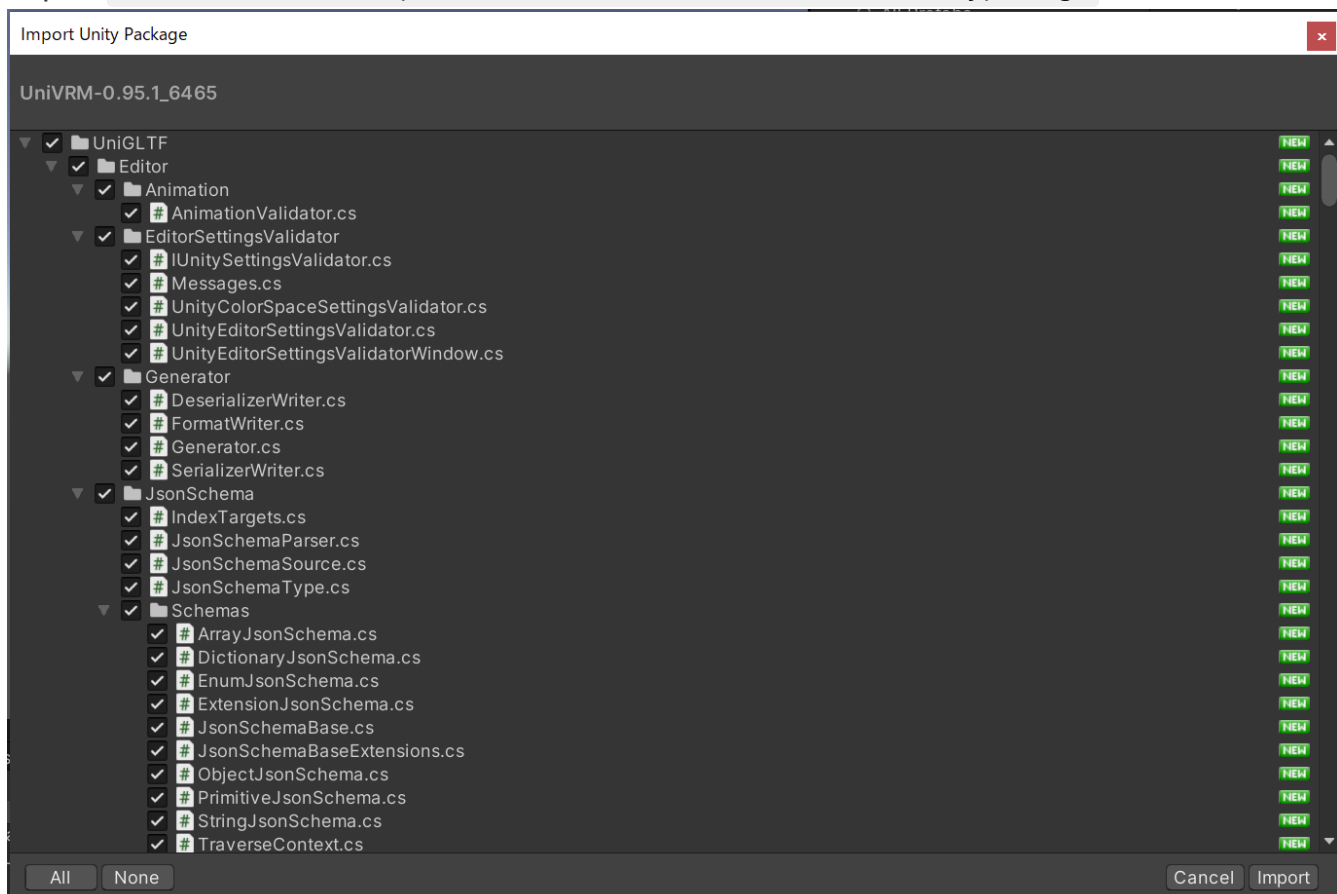




In this example, a model set up with `UnityChan` format is also available.

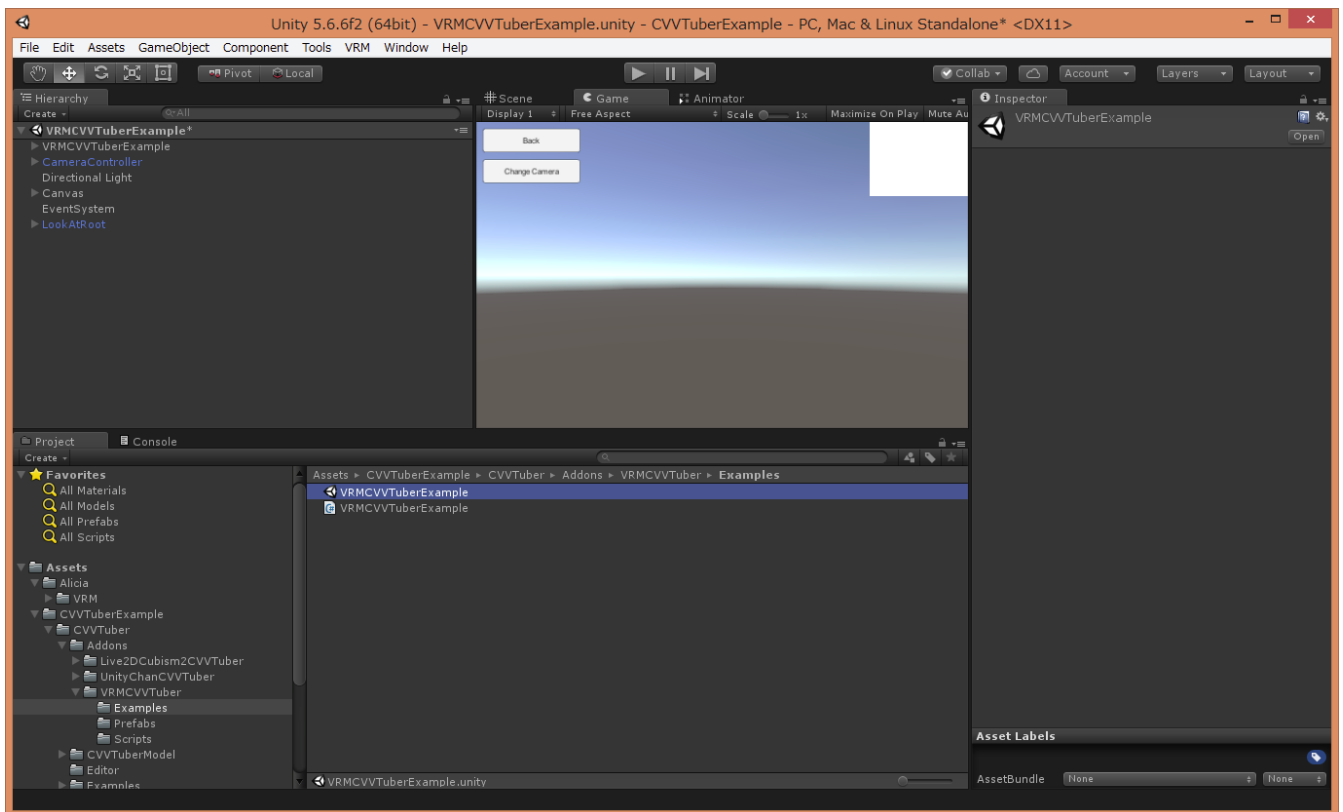
Quick Setup Procedure for VRM10 CV VTuber Example

1. Download `VRM-0.131.2_a471.unitypackage` from [GitHub vrm-c/UniVRM](#).
2. Import `VRM-0.131.2_a471.unitypackage`.
3. Import `Assets/CVVTuberExample/CVVTuber/Addons/VRM10CVVTuber.unitypackage`.

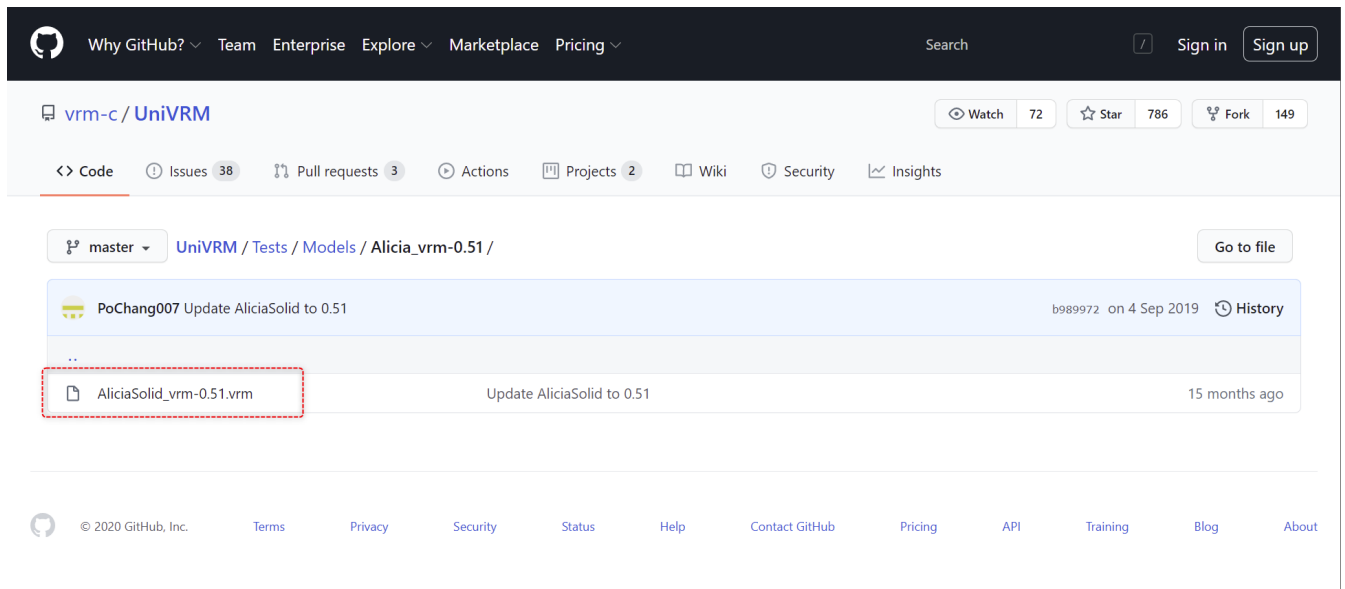


4. Open

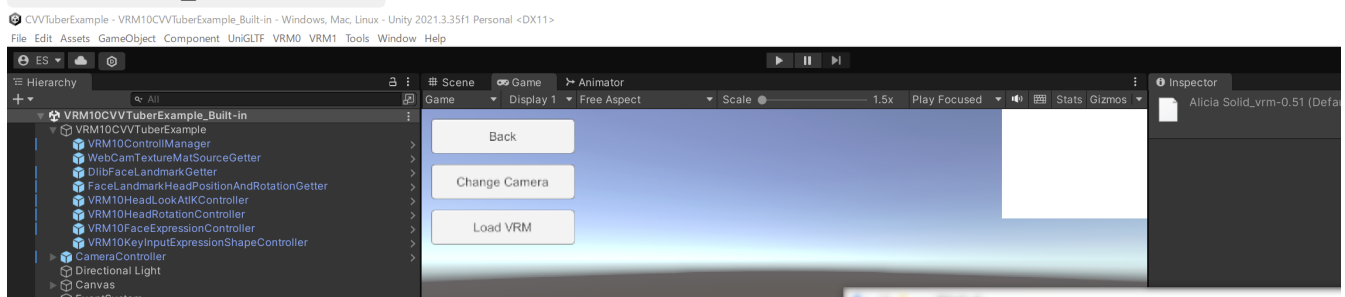
Assets/CVVTuberExample/CVVTuber/Addons/VRM10CVVTuber/Examples/VRM10CVVTuberExample.unity scene.

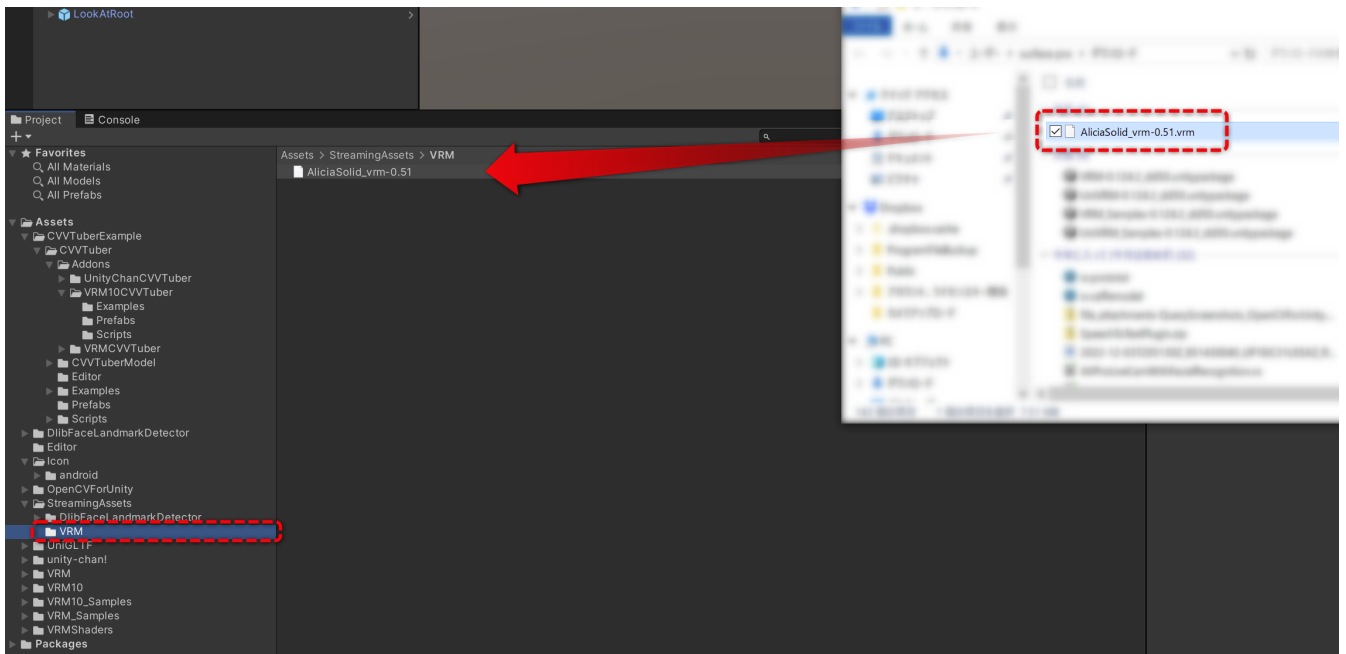


5. Download AliciaSolid_vrm-0.51.vrm from [GitHub vrm-c/UniVRM/Tests/Models/Alicia_vrm-0.51/](https://github.com/vrm-c/UniVRM/Tests/Models/Alicia_vrm-0.51/).

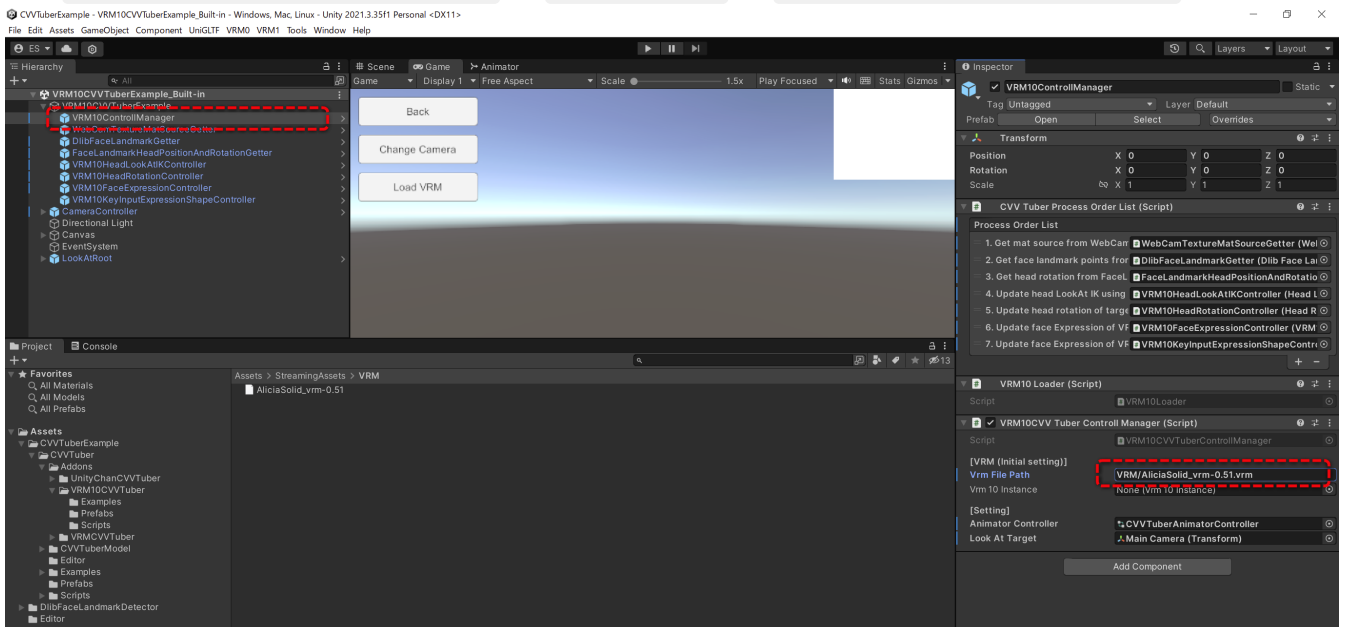


6. Create an VRM folder in the Assets/StreamingAssets folder and place AliciaSolid_vrm-0.51.vrm .

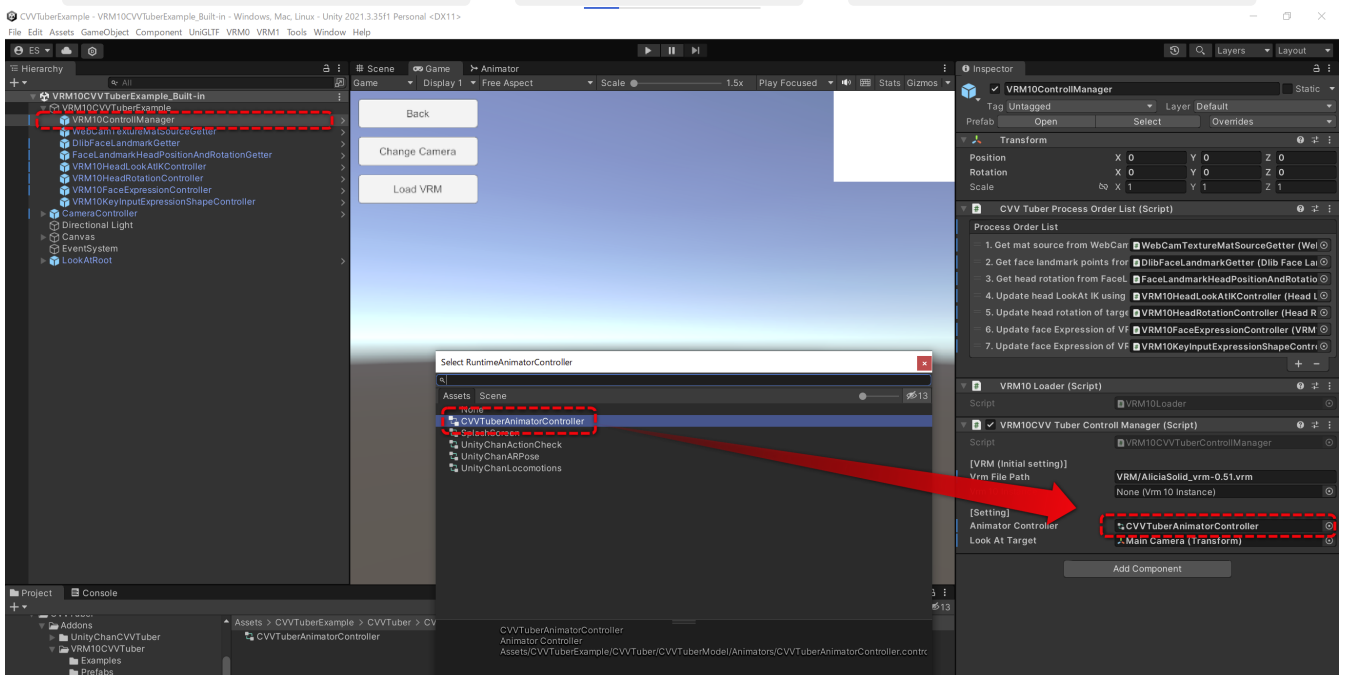




7. Set `VRM/AliciaSolid_vrm-0.51.vrm` to `VrmFilePath` of `VRM10CVVTuberControlManager`.

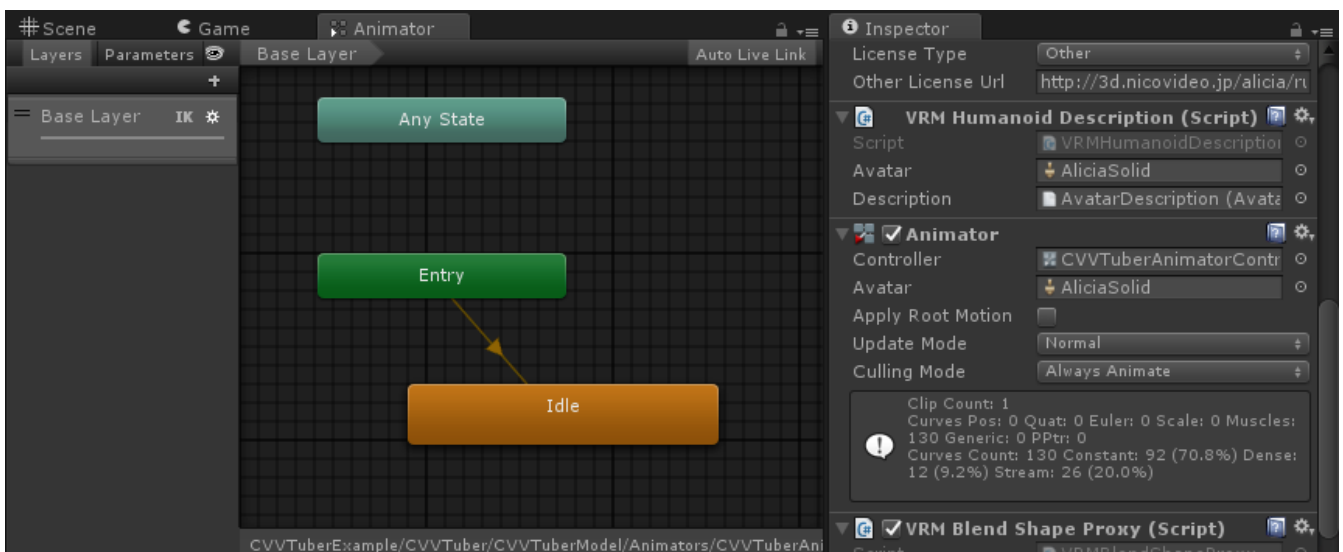
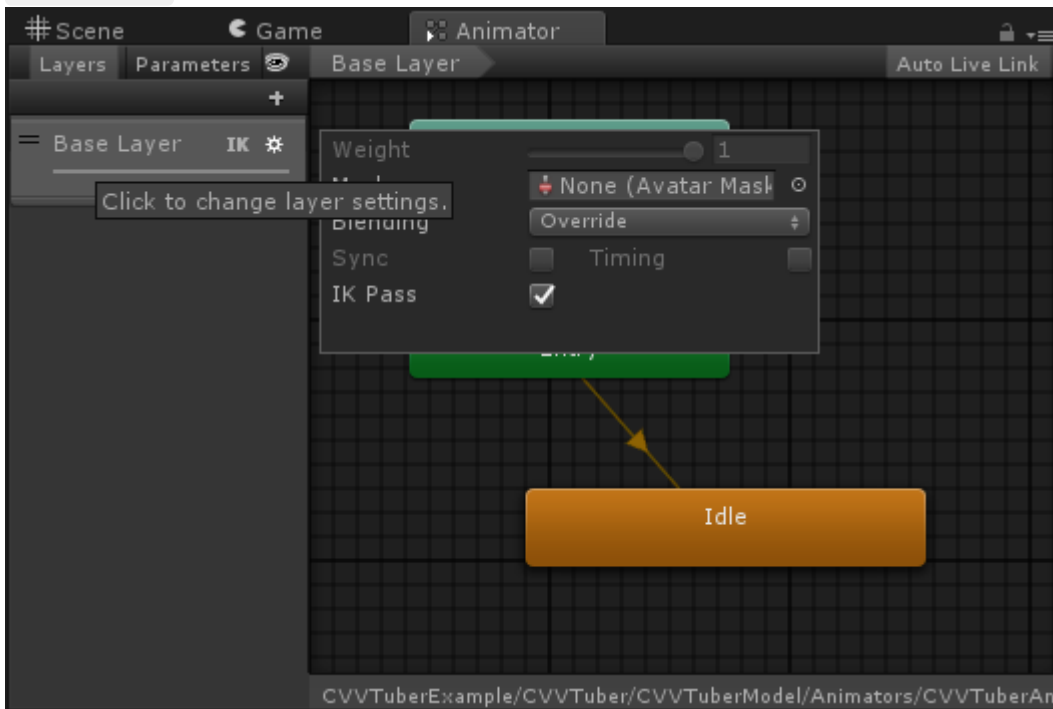


8. Set `CVVTuberAnimatorController` to `AnimatorController` of `VRM10CVVTuberControlManager`.





9. Click on `CVVTuberAnimatorController` to open the `Animator` window. Enable `IK Pass` flag of `Base Layer`.



In this example, a model set up with `VRM` format (1.x) is also available.

`VRM-0.131.2_a471` is compatible with Unity 2022.3 LTS or later.

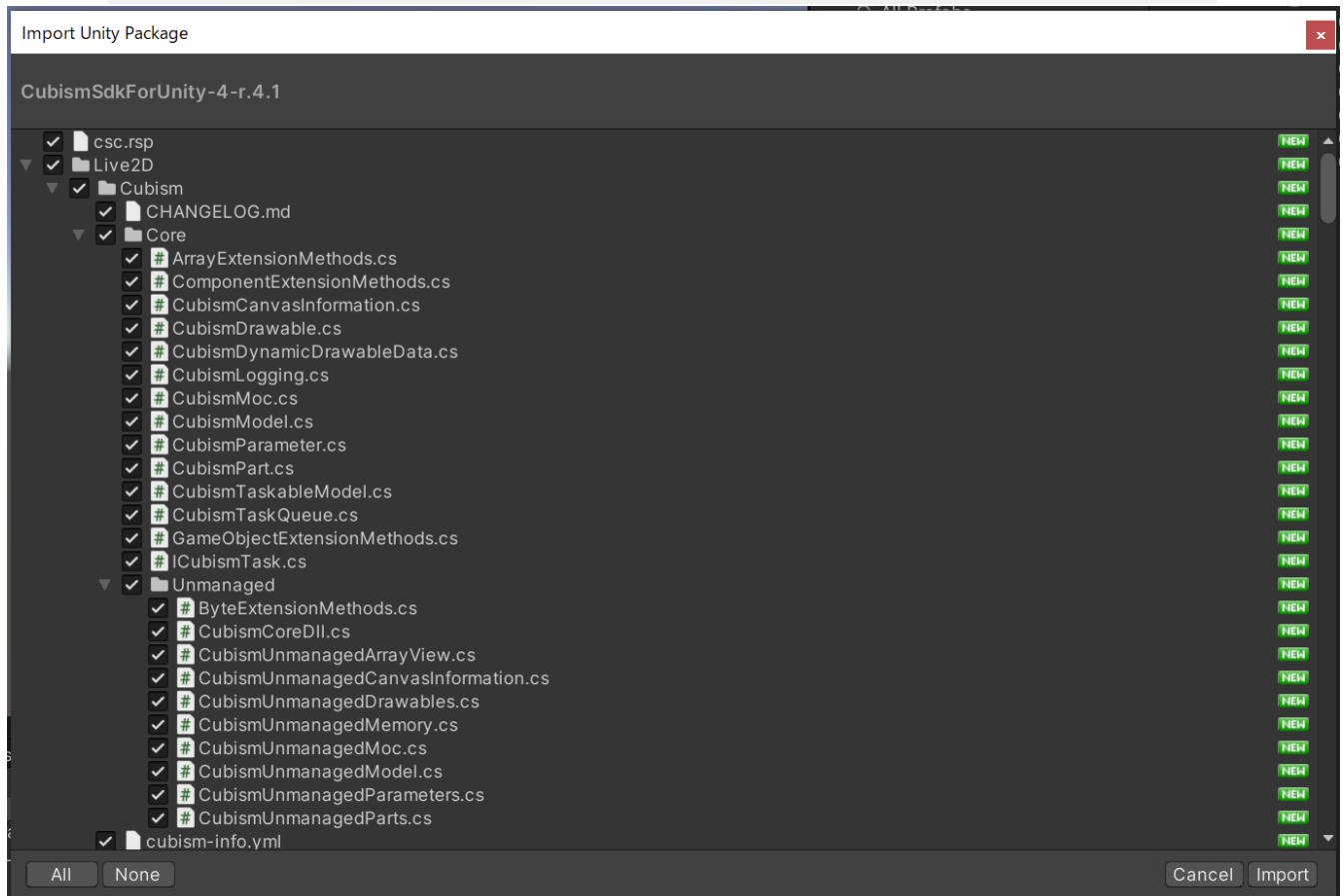
The `Load VRM` button is only enabled on editor runs where the file selection dialog is available. To enable this feature on other platforms, consider adding a plugin such as `UnityStandaLoneFileBrowser`.

Quick Setup Procedure for Live2DCubism5 CV VTuber Example

1. Download `CubismSdkForUnity-5-r.4.2.unitypackage` from [Live2D site](#).

2. Import CubismSdkForUnity-5-r.4.2.unitypackage .

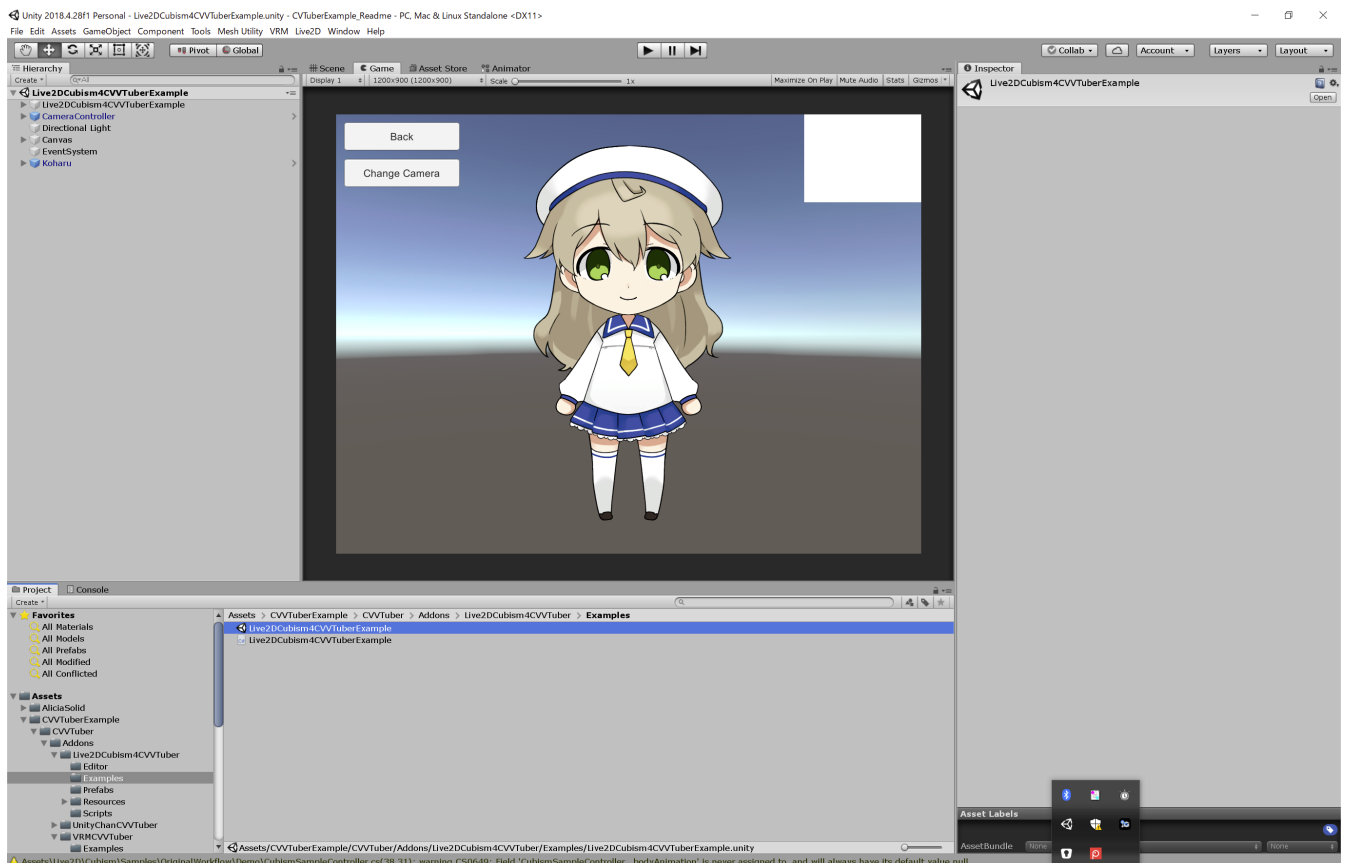
3. Import Assets/CVVTuberExample/CVVTuber/Addons/Live2DCubism5CVVTuber.unitypackage .



4. Open

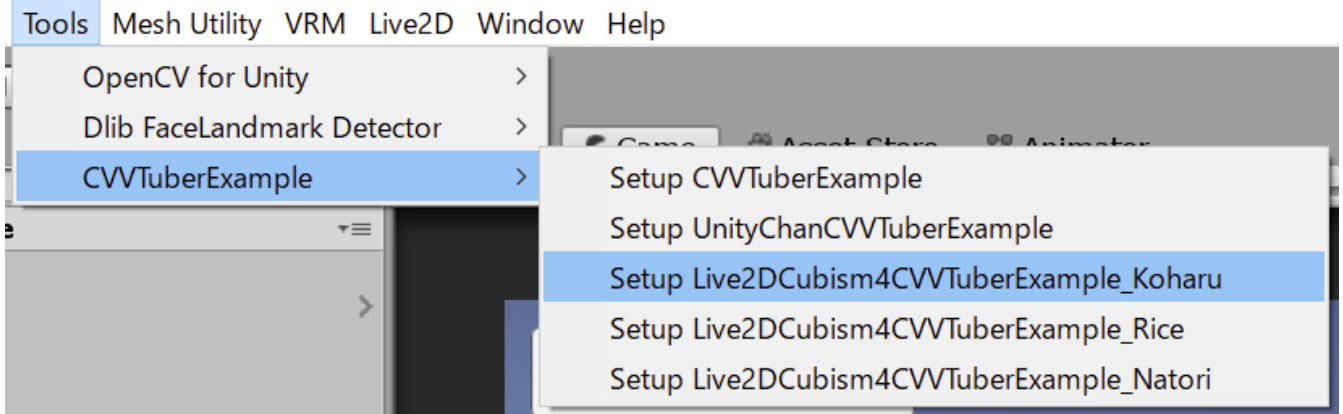
Assets/CVVTuberExample/CVVTuber/Addons/Live2DCubism5CVVTuber/Live2DCubism5CVVTuberExample

scene.



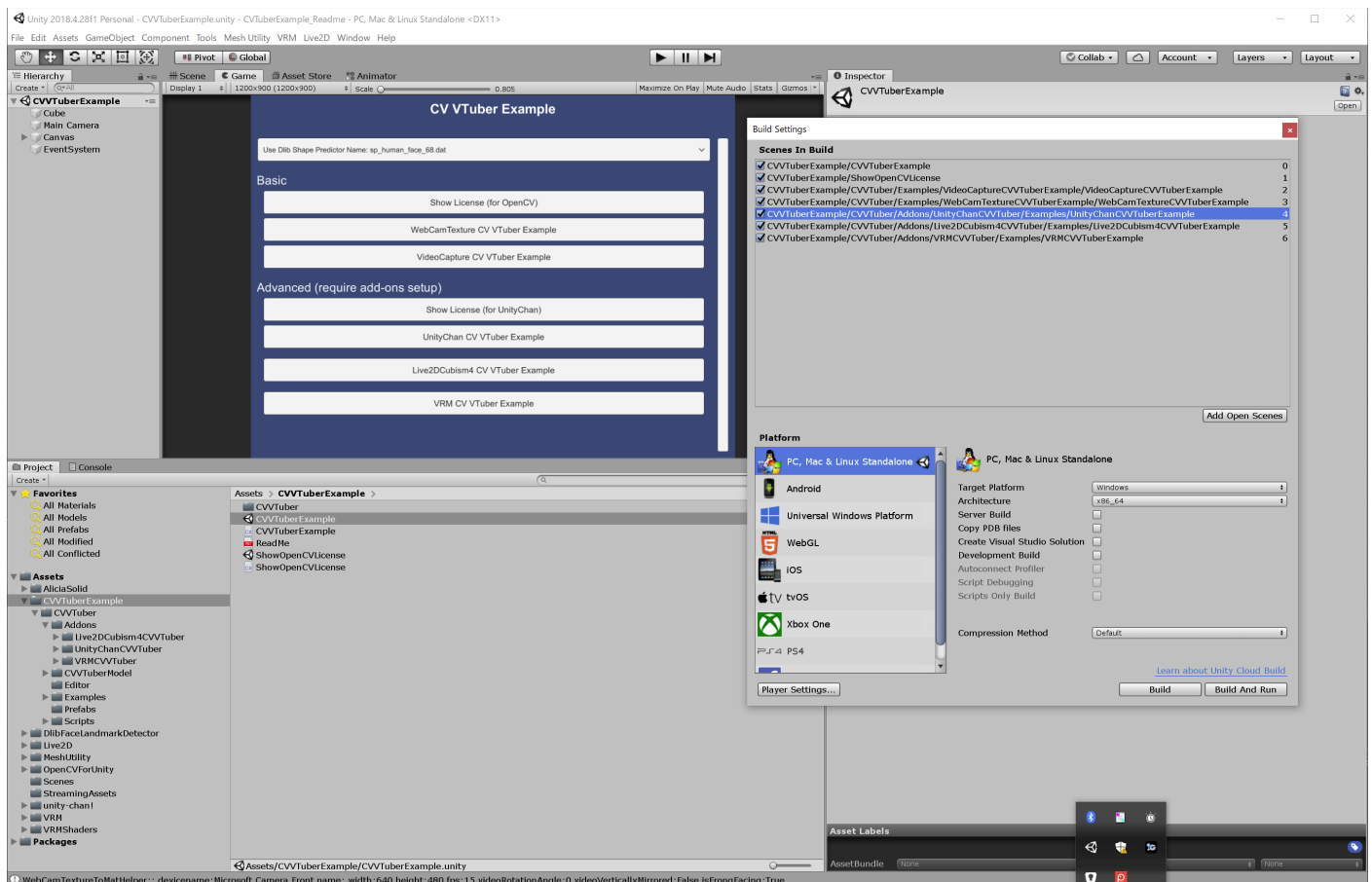
5. Select the menu item

Tools > CVVTuberExample > Setup Live2DCubism5CVVTuberExample_koharu .



In this example, a model set up with `Live2DCubism5` format is also available.
`Cubism 5 SDK for Unity` is compatible with Unity version 2021 LTS or later.

Screenshot after the Setup all addons



? Q & A

Q1. `HeadLookAtIkController` does not work.

A1. Animator is not set to target of HeadLookAtIkController , or IK Pass of Base Layer of AnimatorController is not set to true. Or AnimatorController is not set in the model's Animator .

Q2. HeadRotationController does not work.

A2. HeadRotationController target is not set. (Usually, set the Bone of the Head part)

Q3. The direction of rotation of HeadLookAtIkController or HeadRotationController is wrong.

A3. Please adjust invertAxis and rotateAxis settings.

Version Changes

1.1.4 [Common] Changed the minimum supported version to Unity 6000.0.83.f1. [Common] Updated for CubismSdkForUnity-5-r.4.2.unitypackage. [Common] Updated for VRM-0.131.2_a471.unitypackage. [Common] Removed VRMCVVTuber add-on. [Common] Updated for OpenCV for Unity v3.0.4. (This asset requires OpenCVforUnity 3.0.4 or later.) [Common] Updated for Dlib FaceLandmark Detector v2.0.3. (This asset requires Dlib FaceLandmark Detector v2.0.3 or later.)

1.1.3 [Common] Updated for OpenCV for Unity v3.0.0. (This asset requires OpenCVforUnity 3.0.0 or later.) [Common] Updated for Dlib FaceLandmark Detector v2.0.0. (This asset requires Dlib FaceLandmark Detector v2.0.0 or later.)

1.1.2 [Common] Updated for OpenCV for Unity v2.6.4. (This asset requires OpenCVforUnity 2.6.4 or later.) [Common] Updated for Dlib FaceLandmark Detector v1.4.1. (This asset requires Dlib FaceLandmark Detector v1.4.1 or later.)

1.1.1 [Common] Added VRM10CVVTuberExample. (Compatible with VRM1.0. Depends on VRM-0.124.2_dd50.unitypackage or later.)

1.1.0 [Common] Changed the minimum supported version to Unity2021.3.35f1. [Common] Separated the examples using the Built-in Render Pipeline and Scriptable Render Pipeline. (Except UnityChanCVVTuberExample and VRMCVVTuberExample.) [Common] Updated for UniVRM-0.124.1_e606.unitypackage. Added RuntimeVRMMetaLoader functionality. [Common] Updated for CubismSdkForUnity-5-r.2.unitypackage.

1.0.9 [Common] Updated for OpenCV for Unity v2.5.9. (This asset requires OpenCVforUnity 2.5.9 or later.) [Common] Updated for Dlib FaceLandmark Detector v1.3.8. (This asset requires Dlib FaceLandmark Detector v1.3.8 or later.)

1.0.8 [Common] Updated for OpenCV for Unity v2.5.0. (This asset requires OpenCVforUnity 2.5.0 or later.) [Common] Updated for Dlib FaceLandmark Detector v1.3.4. (This asset requires Dlib

FaceLandmark Detector v1.3.4 or later.) [Common] Updated for UniVRM-0.99.4_8d33. [Common] Updated for Cubism 4 SDK for Unity R5.

1.0.7 [Common] Updated support version to Unity-2019.4LTS. [Common] Updated for OpenCV for Unity v2.4.7. [Common] Updated for Dlib FaceLandmark Detector v1.3.3. [Common] Updated for UniVRM-0.95.1. [Common] Updated for Cubism SDK for Unity R4_1. [Common] Refactored the script.

1.0.6 [Common] Updated for OpenCV for Unity v2.4.2. [Common] Updated for Dlib FaceLandmark Detector v1.3.2. [Common] Removed Live2DCubism2CVVTuber Example and Live2DCubism3CVVTuber Example. [Common] Added Live2DCubism4CVVTuber Example. [Common] Refactored the script.

1.0.5 [Common] Updated for OpenCV for Unity v2.3.8.

1.0.4 [Common] Updated for OpenCV for Unity v2.3.4. [Common] Added Live2DCubism3CVVTuber Example.

1.0.3 [Common] Updated for Dlib FaceLandmark Detector v1.2.6. [Common] Refactored the script.

1.0.2 [Common] Updated for OpenCV for Unity v2.3.3. (This asset requires OpenCVforUnity 2.3.3 or later.) [Common] Updated for Dlib FaceLandmark Detector v1.2.5. (This asset requires Dlib FaceLandmark Detector 1.2.5 or later.)

1.0.1 [Common] Largely changed the folder structure of asset package. (If there is a previous version of CVVTuberExample in the project, please delete the CVVTuberExample folder first and then import the new version.) [Common] Added WebCamTextureCVVTuberExample, VideoCaptureCVVTuberExample, UnityChanCVVTuberExample, VRMCVVTuberExample and Live2DCubism2CVVTuberExample.

1.0.0 Initial version.